

1. EXECUTIVE PLATFORM ARCHITECTURE & CONSTRUCTION OVERVIEW

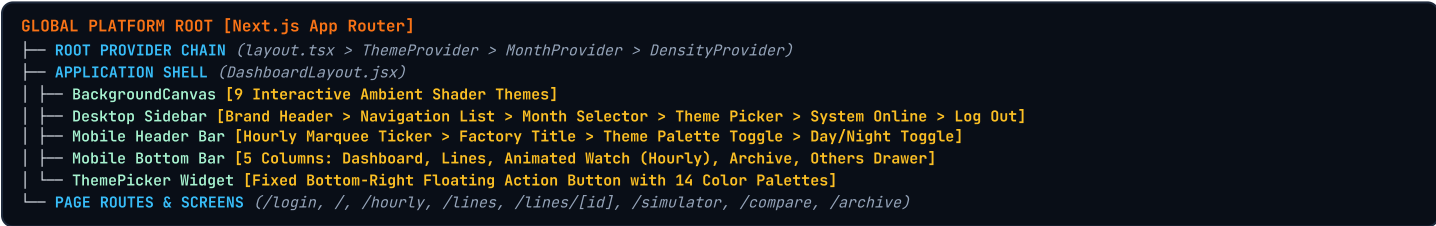
The **Byzid Apparels Factory Operations Platform** is a high-speed telemetry and financial analytics dashboard engineered for industrial garment manufacturing. Built on Next.js 16.2 (Turbopack) and React 19, the website provides floor-to-executive visibility across 4 sewing lines, live hourly piece counts, dynamic Contribution Margin (CM) financial calculations, and interactive visual physics simulations.



2. COMPLETE WEBSITE ANATOMY INDEX (EVERY PAGE & URL ROUTE)

PAGE NAME	URL ROUTE	CORE COMPONENT FILE	PRIMARY SCREEN PURPOSE & KEY FEATURES
Login Portal	/login	src/app/login/page.jsx	Access code security gate, glassmorphic card, passcode visibility toggle, auto-focus input.
Executive Dashboard	/	src/components/dashboard/DashboardContent.jsx	Top Month tabs, Visual Titanic animation, 6 KPI cards, 4-Line summary grid, AI Insights, Telemetry charts.
Hourly Production	/hourly	src/app/hourly/page.jsx	Live floor tracking, calendar date picker, 11-hour matrix (Target vs Actual vs Variance), hourly DHU % defects.
Lines Overview	/lines	src/app/lines/page.jsx	Single-row Month Switcher + Density Switcher (Compact table, Normal cards, Detailed telemetry), floor stats.
Line Telemetry Deep Dive	/lines/[id]	src/components/dashboard/LineDetailsContent.jsx	Dedicated Line A, B, C, D analytics, breadcrumb trail, 6-box Operations Summary, daily historical logs.
Simulation Studio	/simulator	src/app/simulator/page.jsx	Interactive physics simulator, Titanic floating/sinking engine, Profit/Neutral/Loss modes, 1-31 day loss slider.
Compare Workbench	/compare	src/app/compare/page.jsx	Benchmarking workbench: Line vs Line, Date vs Date, Month vs Month with side-by-side radar and bar charts.
Archival Catalog	/archive	src/app/archive/page.jsx	Catalog of past months (July, August, September), financial status badges, external Google Sheets download links.
Printable Report Engine	/report	src/components/report/PrintableArchiveReport.jsx	A4 print-ready executive summary generator, breakdown of floor operational costs and production earnings.

3. MASTER CONSTRUCTION HIERARCHY TREE (TOP-LEVEL STRUCTURE)



4. GLOBAL LAYOUT SHELL CONSTRUCTION & COMPONENT BREAKDOWN

GLOBAL SHELL > DASHBOARD.LAYOUT.JSX

- 1. BACKGROUND CANVAS (`<BackgroundCanvas />`)
 - Theme Shaders: Ambient Orbs, Ember Tide Aurora, Cyber Grid, Abstract Void, Nebula Pulse, Minimal Slate
 - Interactive Pointer: Radial mouse spotlight cursor tracker on desktop screens
- 2. DESKTOP SIDEBAR (`<Sidebar />` - Fixed Left Navigation, Collapsible 240px to 72px)
 - Brand Header: Byzid Apparels Logo icon, Factory title, Version badge (v8.0.9)
 - Primary Nav Links: Dashboard (`/`), Hourly (`/hourly`), Lines (`/lines`), Compare (`/compare`), Analytics, Simulator, Archive, Byzid Profile (`external`)
 - Nested Lines Dropdown: Sub-links to Line A (Green), Line B (Blue), Line C (Purple), Line D (Amber)
 - Month Selector Radio: September (LIVE), August, July active state switcher
 - Sidebar Theme Palette Accordion: Color palette circles (14 themes) + Background shader selector + Night/Day mode switch
 - System Online Widget: Green pulse ping status indicator + Last Updated timestamp
 - Log Out Button: Red border button, executes DELETE `/api/auth` and redirects to `/login`
- 3. MOBILE HEADER BAR (*Visible on screen width < 768px*)
 - Live Hourly Marquee Ticker: Top scrolling ribbon with latest hour piece output per line
 - Brand Identity: "BYZID APPARELS PVT LTD" title + "Last Updated: [Date]"
 - Theme Palette Trigger: Opens mobile appearance & style drawer modal
 - Day / Night Toggle Button: Direct one-tap switch between Dark mode and Light mode
- 4. MOBILE BOTTOM NAVIGATION BAR (`<MobileBottomNav />` - Fixed Bottom Floating Pill)
 - Col 1 - Dashboard: Home icon, active primary highlight
 - Col 2 - Lines: Factory icon, routes to `/lines` overview
 - Col 3 - Center Animated Watch (Hourly): Elevated glowing circular button with rotating watch hands, routes to `/hourly`
 - Col 4 - Archive: History icon, routes to `/archive` catalog
 - Col 5 - Others Drawer Trigger: Three-dots horizontal icon, triggers popup drawer
- 5. MOBILE OTHERS DRAWER POPUP (*Slides up from bottom right*)
 - Canteen Trigger: Opens Canteen Security & PIN Verification Modal
 - Internal Tools: Simulator, Analytics, Compare Workbench
 - Direct Line Jumpers: Line A, Line B, Line C, Line D color-coded pills
 - External Link: Byzid Corporate Profile (Vercel)
 - Mobile Log Out Action: Instant session clearance and redirection to login screen
- 6. FLOATING THEME PICKER (`<ThemePicker />` - Fixed bottom-right action button)
 - Trigger: 40px circular palette button with glowing halo
 - Flyout Popup: 14 color themes (Oceanic, Ember Tide, Emerald, Rose, etc.), Dark/Light switcher, Log Out button

5. AUTHENTICATION GATEWAY & PRE-AUTH LOGIN SCREEN (/LOGIN)

Login Portal Anatomy (/login)

The authentication portal is an isolated, high-security pre-auth route with custom industrial aesthetics:

LOGIN SCREEN ANATOMY

- Industrial Background: Architectural blueprint grid, security status watermarks, radial spotlight
- Glassmorphic Card: Elevated matte container with glowing borders
- Apparel Crest: Factory vector badge with sewing needle & shield emblem
- Portal Title: "BYZID APPARELS" & "Operations Access Portal"
- Passcode Input: Password field with Eye/EyeOff visibility toggle, auto-focused
- Unlock Button: "AUTHENTICATE" action with arrow animation and loading spinner
- Footer Notice: "Protected Facility Portal · Authorized Factory Management Only"

Security & Loop-Prevention Implementation

Edge middleware protects all routes. Background polling aborts immediately if 401 Unauthorized occurs:

```
// Edge Route Proxy Gateway (src/proxy.js)
export function middleware(req) {
  const session = req.cookies.get('ba_session')?.value;
  const isAuthenticated = session === hashPasscode(process.env.DASHBOARD_PASSCODE);
  if (!isAuthenticated && !req.nextUrl.pathname.startsWith('/login')) {
    return NextResponse.redirect(new URL('/login', req.url));
  }
}

// 401 Flash Prevention (useKpiData.js & ClientAutoRefresh.jsx)
if (res.status === 401) {
  if (window.location.pathname === '/login') {
    window.location.href = '/login';
  }
  return null; // Stop polling loop completely
}
```

6. EXECUTIVE DASHBOARD (/) HIERARCHICAL BREAKDOWN

The primary operational screen combines executive-level financial intelligence with floor production analytics. The breakdown below details every nested section, button, and widget:

```
DASHBOARD SCREEN ROOT [src/app/page.jsx > DashboardContent.jsx]
├─ SECTION 1: TOP CONTROLS & VISUALS TOGGLE
│  └─ Titanic Animation (Visual Sinking Voyage) [TitanicAnimation.jsx]
│     └─ Dynamic Vessel Model: High-detail SVG luxury ocean liner with smoking funnels and flag
│     └─ Physics Engine: Ship tilts and sinks into animated SVG water waves if Net Profit is negative
│     └─ Floating Full Steam Ahead: Ship sails smoothly forward when month is profitable
│     └─ Water Level Meter: Dynamically calculates water breach line according to cumulative loss
│     └─ Visuals Toggle Button ("Show Visuals" / "Hide Visuals")
│        └─ Pill button centered above header; features pulsing slow-blink animation when visuals are collapsed
├─ Factory Financials Header
│  └─ Title: "FACTORY FINANCIALS" (Uppercase gradient text with shadow)
│  └─ Analyst Credit: "Financial Data Analyst : Rofiqul Islam Zia (Merchandiser)"
├─ SECTION 2: 6 KPI METRIC STAT CARDS [MetricCard.jsx × 6]
│  └─ Month Switcher Tabs: Pill switch for September (LIVE), August, and July
│  └─ Date Range Subtitle: Dynamic badge e.g. FROM 01 SEP TO 09 SEP, 2026
│  └─ Card 1: Total Cost (BDT): Sum of operator labor, overhead, and operating expenses + % vs previous month
│  └─ Card 2: Total Income (BDT): Cumulative CM gross earnings generated from all production lines
│  └─ Card 3: Net Profit / Loss (BDT): Net financial balance; dynamically colored (Green = Profit, Red = Loss)
│  └─ Card 4: Working Days: Total recorded active manufacturing days in selected month (e.g. 8 Days)
│  └─ Card 5: Production Lines: Active producing lines indicator (e.g. "All 4 Lines Active & Producing")
│  └─ Card 6: Avg Daily Profit / Loss (BDT): Daily run-rate deficit or profit pace (e.g. "-109k / day")
├─ SECTION 3: PRODUCTION LINE SUMMARY SECTION
│  └─ Section Header: "LINE SUMMARY SEPTEMBER" + Date range badge
│  └─ Mobile Line Switcher Tabs: 4 single-row buttons: [LINE A] [LINE B] [LINE C] [LINE D]
│  └─ Desktop 2x2 Grid Layout: Line A & B in top row, Line C & D in bottom row
│  └─ Inside Each Line Card (Component Structure):
│     └─ Color Top Accent Bar: Emerald (A), Royal Blue (B), Deep Purple (C), Warm Amber (D)
│     └─ Line Title & Status Badge: Line Name + Status (Optimal, Idle, or Critical with strobe beacon)
│     └─ Active Item & Efficiency: Running garment (e.g. Sherpa Jacket, Boxer) + Cost Recovery Efficiency %
│     └─ Month Total Summary (4-box row): Production (PCS), Income (BDT), Cost (BDT), Net Profit/Loss (BDT)
│     └─ Last Day Input Strip (3-box row): Day Output (PCS), Day Income (BDT), Day Net Profit/Loss (BDT)
│     └─ Action CTA Button: "VIEW FULL DETAILS" button routing directly to /lines/[id]
├─ SECTION 4: SYSTEM INTELLIGENCE (AI INSIGHTS PANEL)
│  └─ AI Sparkles Icon: Illuminated accent container
│  └─ Automated Bullet Insights: Dynamic operational summary generated from floor telemetry rules
├─ SECTION 5: PERFORMANCE TELEMETRY CHARTS
│  └─ Daily Output Area Chart: Recharts area visualization plotting daily Target vs Actual pieces produced
│  └─ Income vs Cost Bar Chart: Bi-color bar graph comparing daily financial gross revenue against factory expenses
```

7. DASHBOARD LINE CARDS TECHNICAL DATA MATRIX

LINE ID	ACCENT COLOR	RUNNING ITEM	TARGET MANPOWER	PRIMARY BUYER	DEFAULT TARGET / HOUR
Line A	#10B981 (EMERALD)	SHERPA JACKET / FLANNEL	78 Operators	FAMETEX	80 Pcs / Hour
Line B	#3B82F6 (BLUE)	BOXER SHORTS	48 Operators	TOP-GRADE	230 Pcs / Hour
Line C	#A855F7 (PURPLE)	BOXER SHORTS	48 Operators	TOP-GRADE	230 Pcs / Hour
Line D	#F59E0B (AMBER)	MEN'S T-SHIRT (STANDBY)	45 Operators	INTERNAL	Standby / Idle

8. HOURLY PRODUCTION MATRIX (/HOURLY) HIERARCHICAL BREAKDOWN

```
HOURLY FLOOR TRACKING ROOT [src/app/hourly/page.jsx]
├─ TOP FLOOR CONTROLS & HEADER
│  └─ Page Title: "HOURLY PRODUCTION" + Subtitle "REAL-TIME FLOOR PRODUCTION LOG"
│  └─ Live Sync Status Indicator: Real-time clock (<RealTimeClock />) + pulsating green sync dot
│  └─ Date Selector Dropdown: Interactive calendar selector popup listing all available production dates
│  └─ Date Row: Date label, today's badge, Friday factory rest badge, or public holiday banner
│  └─ Total Pieces Indicator: Shows total finished pieces produced on that date directly inside dropdown
├─ FACTORY TOTAL SUMMARY STRIP
│  └─ Total Target Box: Cumulative target sum for all 4 lines combined
│  └─ Total Actual Box: Cumulative actual pieces sewn across the entire factory
│  └─ Overall Achievement Gauge: Animated precision radial ring (<PrecisionProgressRing />) showing %
│  └─ Active Shift Indicator: Automatically detects currently running hour (e.g. "8th Hour Active")
├─ DESKTOP 11-HOUR PRODUCTION MATRIX TABLE
│  └─ Sticky Left Column: Line Identifiers (Line A, B, C, D) + Combined Factory Total Row
│  └─ 11 Hourly Shift Columns: 1ST, 2ND, 3RD, 4TH, 5TH, 6TH, 7TH, 8TH, 9TH, 10TH, 11TH Hours
│  └─ Cell Data Triad: Target pieces, Actual pieces, and Achievement % in every hourly cell
│  └─ Cell Note Callouts: Operator downtime or style-change notes rendered under cell value
├─ MOBILE HOURLY FLOOR EXPERIENCE
└─ Single-Row Mobile Tab Bar: [OVERALL] [LINE A] [LINE B] [LINE C] [LINE D]
└─ Overall Tab: Compact 4-column table (Hour, Target, Actual, Achieve %)
└─ Line Tabs: Line Specs Grid (Buyer, Style, Item, Workers) + 11-hour vertical breakdown table + Total Footer
```

9. LINES OVERVIEW (/LINES) CONSTRUCTION & DENSITY ENGINE

```
LINES OVERVIEW ROOT [src/app/lines/page.jsx]
├─ CONTROLS STRIP (Enforced Single-Row Flex-Nowrap)
│  └─ Month Switcher Tabs: September (LIVE with red badge), August, July
│  └─ Density Switcher (<DensitySwitcher />): Single-row 3-way toggle button:
│     └─ COMPACT (Table icon) · Condensed spreadsheet table for high-speed executive scanning
│     └─ NORMAL (Grid icon) · Balanced card grid with key performance badges and CTAs
│     └─ DETAILED (List icon) · Expanded engineering telemetry with hourly trend metrics
├─ FACTORY FLOOR SUMMARY STRIP
│  └─ 4 stat cards: Total Target (PCS), Total Actual (PCS), Total Income (BDT), Net Profit / Loss (BDT)
├─ VIEW MODE 1: COMPACT EXECUTIVE SPREADSHEET (Density === 'compact')
│  └─ High-density table: Line, Active Item, Workers, Month Output, Total Income, Total Cost, Net Profit, Recovery %, Status, Action
├─ VIEW MODE 2: BALANCED CARD GRID (Density === 'normal')
│  └─ 4 responsive line cards: Top accent bar, Worker counts, Efficiency gauge, Last day input stats, "View Details" button
├─ VIEW MODE 3: DETAILED TELEMETRY (Density === 'detailed')
└─ Expanded line cards with inline break-even financial gauges and worker allocation graphs
```

10. COMPARISON OF DENSITY MODES IN LINES OVERVIEW

DENSITY MODE	LAYOUT FORMAT	TARGET SCREEN	VISIBLE DATA FIELDS
Compact	Single Responsive Table	Desktop & Mobile Tablets	Line ID, Item, Workers, Pcs Output, Total Income, Total Cost, Net Profit, Cost Recovery %, Status Pill.
Normal	2x2 Grid of Cards	Standard Mobile & Desktop	Line identity, Buyer, Item, Efficiency %, Month totals (4 stats), Last day input (3 stats), Navigation CTA.
Detailed	Expanded Modular Panels	Industrial Floor Monitors	Full engineering telemetry, operator counts, break-even graphs, daily CM contribution analysis.

11. LINE TELEMETRY DEEP DIVE (/LINES/[ID]) CONSTRUCTION

LINE DEEP DIVE ROOT [LineDetailsContent.jsx]

- HEADER & BREADCRUMB BAR
 - Breadcrumbs: Dashboard > Lines > Line [A/B/C/D] > Telemetry
 - Back Navigation: Circle arrow button returning to previous view
 - Title & Subtitle: "Line [ID] Telemetry" + "Detailed breakdown synced from source records"
 - Local Density Switcher: Independent Compact / Normal / Detailed table toggle
- OPERATIONS SUMMARY GRID (6-Card Container)
 - Date Range: Active date range (e.g. "01 - 09 Sep, 2026")
 - Active Item: Garment style currently running on line (e.g. "SHERPA JACKET")
 - Total Production: Cumulative finished pieces sewn in selected period
 - Total Cost: Cumulative line operational expenses in BDT
 - Total Income: Cumulative CM earnings generated by line in BDT
 - Net Balance: Net Profit / Net Loss with green/red status glow
- DAILY HISTORICAL LOGS TABLE
 - Table Columns: Date, Line, Style, Item, Workers, Per Head Cost, Total Cost, Prod Qty, Prod DZN, CM/DZN, Total Income, Net Profit
 - Special Row Handlers: Friday Factory Rest rows and Public Holiday rows automatically highlighted in amber/rose

12. SIMULATION STUDIO (/SIMULATOR) CONSTRUCTION & CONTROLS

SIMULATION STUDIO ROOT [src/app/simulator/page.jsx]

- 1. HOW THE SIMULATOR WORKS (Interactive Accordion)
 - Accordion Header: Question mark icon + Title + Expand/Collapse toggle [+ / -]
 - Explanation Body: Details Profitable (full steam), Breaking Even (neutral), and Sinking (loss progression)
- 2. SIMULATED TITANIC ANIMATION STAGE
 - Live SVG ship stage responding directly to simulation state variables (simMode, simDay)
- 3. SHIP PHYSICS CONTROL PANEL
 - Mode Toggles (3 Buttons):
 - PROFIT MODE (Green button): Ship sails forward with full steam, waves ripple smoothly
 - NEUTRAL MODE (Amber button): Ship stays afloat at neutral waterline
 - LOSS MODE (Red button): Triggers water breach physics and unlocks time progression slider
 - Time Progression Slider (Only active in Loss Mode):
 - Range: Day 1 to Day 31 of sustained losses
 - Visual Milestones: Safe → Water Breach → 30° Stern Tilt → 90° Vertical Plunge → Vessel Shatter

13. COMPARE WORKBENCH (/COMPARE) CONSTRUCTION

COMPARE WORKBENCH ROOT [src/app/compare/page.jsx]

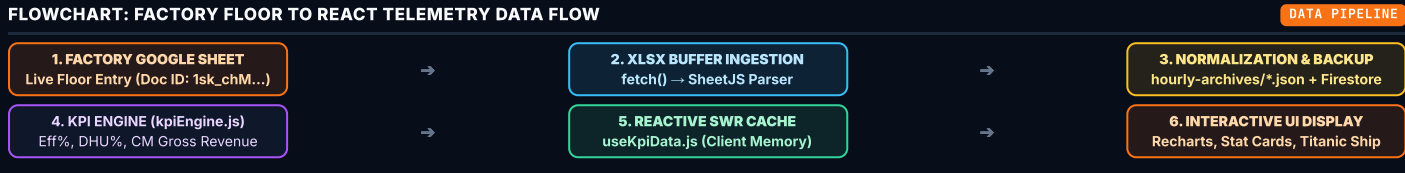
- TAB 1: LINE COMPARISON (Line vs Line)
 - Selectors: Line 1 Selector (A, B, C, D) vs Line 2 Selector (A, B, C, D)
 - Side-by-Side KPI Cards: Production Output, Efficiency %, Total Revenue, Cost, Profit Delta
 - Comparative Charts: Multi-axis Radar comparison chart & Bar delta comparison graph
- TAB 2: DATE COMPARISON (Day vs Day)
 - Selectors: Date 1 dropdown vs Date 2 dropdown
 - Hourly Curve Overlay: Recharts line graph overlaying both days' 11-hour output curves
- TAB 3: MONTH COMPARISON (Month vs Month)
 - Selectors: Month 1 dropdown (September LIVE) vs Month 2 (August / July)
 - Financial Variance Analysis: Month-over-month CM variance, working day counts, and daily average loss comparisons

14. ARCHIVAL CATALOG (/ARCHIVE) & REPORT GENERATOR (/REPORT)

ARCHIVE & REPORTS ROOT [src/app/archive/page.jsx]

- MONTHLY ARCHIVE DIRECTORY
 - July 2026 Card: Recorded working days, net balance, "SEE DETAILS" action → /archive/2026-07
 - August 2026 Card: Recorded working days, net balance, "SEE DETAILS" action → /archive/2026-08
 - September 2026 Card: Live month snapshot → /
- EXTERNAL GOOGLE DRIVE PDF SHEET DOWNLOADS
 - July Factory Sheet: Direct Google Drive link (1GWGOeDtG3mvxW7T8y0w64UHG75jBR-Bv)
 - August Factory Sheet: Direct Google Drive link (1-Jug6L8J5_CttfrAyrh2NCFuV70g0pvu)
- PRINTABLE EXECUTIVE REPORT GENERATOR [PrintableArchiveReport.jsx]
 - Generates clean, printer-optimized A4 financial audits formatted for factory board meetings

15. COMPLETE DATA INGESTION & CALCULATION PIPELINE FLOWCHART



16. AI PLATFORM COLLABORATION PROMPT & SYSTEM RULES

Ready-to-Use Prompt Template for Sharing with External AI Models:

Copy and paste the exact prompt below into ChatGPT, Claude, Gemini, or other AI platforms to provide instant context:

"CONTEXT: You are assisting with the Byzid Apparels Factory Operations Platform (Byzid Apparels Pvt Ltd).

- Framework: Next.js 16.2 Turbopack, React 19, Tailwind CSS v4, Capacitor 8.5 (Android native).
- Production Theme: 'ember-tide' in enforced Night/Dark mode (0ms FOUC prevention via blocking script in layout.tsx).
- Security: ALL routes protected by src/proxy.js using SHA-256 session cookie (ba_session). Passcode is 'do1@z1a'. 401 calls abort polling loops to prevent login screen flashing.
- Pages & Construction:
 - Login (/login): Glassmorphic pre-auth portal with passcode visibility toggle.
 - Dashboard (/): Top controls > Show/Hide Visuals > Titanic SVG animation > Factory Financials header > 6 KPI cards > Line Summary (Line A-0 tabs/grid) > AI Insights > Performance charts.
 - Hourly (/hourly): Date picker > RealTimeClock > Total Production card > 11-Hour matrix table (Target vs Actual vs Variance vs Notes).
 - Lines (/lines): Single-row Month Switcher (flex-nowrap) + Density Switcher (Compact table / Normal cards / Detailed telemetry).
 - Line Details (/lines/[id]): Breadcrumb nav > Operations Summary 6-grid > Daily historical logs table.
 - Simulator (/simulator): Titanic floating vs sinking simulator > Profit/Neutral/Loss modes > 1-31 Day Loss slider.
 - Compare (/compare): Line vs Line, Date vs Date, and Month vs Month with Recharts radar and bar graphs.
 - Archive (/archive): Monthly catalog cards with Google Drive PDF links.
- Math Formulas (src/utils/kpiEngine.js):
 - Efficiency % = (Actual Output × SMV) / (Manpower × Working Minutes) × 100
 - Contribution Margin = Actual Pieces × (CM per Dozen / 12) - Daily Factory Overhead

Please respect the component hierarchy and 60-30-10 color budget (Skill.md) in all modifications."